Nutrients & Carbohydrates – Page 421

Source: Lucent's General Knowledge – Biology

Biology: Nutrients, Carbohydrates & Energy Production

50 Questions • 30 Minutes

Instructions:
- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. The function of food in providing basic nutrition is called:

A. Digestion
B. Nutrition
C. Absorption
D. Excretion

Ans: B

Q2. Nutrients include:

A. Only proteins
B. Carbohydrates, proteins, fats, minerals, vitamins and water
C. Only vitamins
D. Only minerals

Ans: B

Q3. Carbohydrates are organic compounds made of:

A. Nitrogen and Hydrogen
B. Carbon, Hydrogen and Oxygen
C. Sulphur and Phosphorus
D. Only Carbon

Ans: B

Q4. Carbohydrate is the major intake in animals and humans, providing what percentage of energy?

A. 20-30%
B. 50 to 75%
C. 90%
D. 10%

Ans: B

Q5. Carbohydrate containing aldehyde group is called:

A. Ketose
B. Aldose
C. Lactose
D. Sucrose

Ans: B

Q6. Carbohydrates are classified into how many main groups?

A. 2
B. 3 – Monosaccharides, Oligosaccharides, Polysaccharides
C. 4
D. 5

Ans: B

Q7. Monosaccharides are the simplest sugars composed of:

A. Multiple units
B. Single polyhydroxy or ketone units
C. Proteins
D. Fats

Ans: B

Q8. Most abundant monosaccharide found in nature is: **A.** Fructose **B.** Glucose **C.** Galactose **D.** Ribose	**Ans: B**
Q9. Glucose is a type of: **A.** Disaccharide **B.** Hexose sugar **C.** Polysaccharide **D.** Amino acid	**Ans: B**
Q10. The sweetest sugar is: **A.** Glucose **B.** Fructose **C.** Sucrose **D.** Lactose	**Ans: B**
Q11. When 2 to 10 monosaccharides join together they form: **A.** Polysaccharides **B.** Oligosaccharides **C.** Amino acids **D.** Fatty acids	**Ans: B**
Q12. Maltose, Sucrose and Lactose are usually: **A.** Monosaccharides **B.** Disaccharides (crystalline, sweet in taste) **C.** Polysaccharides **D.** Proteins	**Ans: B**
Q13. Polysaccharides are formed when more than how many monosaccharides join? **A.** 2 **B.** 10 **C.** 5 **D.** 3	**Ans: B**
Q14. Polysaccharides are large, insoluble, tasteless. Examples include: **A.** Glucose and Fructose **B.** Starch, glycogen, cellulose, chitin **C.** Maltose and Sucrose **D.** Ribose and Deoxyribose	**Ans: B**
Q15. Glucose is the source of immediate energy production in: **A.** Plants only **B.** The cell **C.** Bones **D.** Hair	**Ans: B**

Q16. Carbohydrate is used as fuel. During the process of respiration, glucose breaks into CO2 and H2O with the help of: **A.** Vitamins **B.** Oxygen **C.** Minerals **D.** Hormones	**Ans: B**

Q17. One gram of glucose gives how many kilo calories of energy? **A.** 9.3 **B.** 4.2 **C.** 7.0 **D.** 2.1	**Ans: B**

Q18. Nucleic acids are polymers of: **A.** Amino acids **B.** Nucleosides and nucleotides **C.** Fatty acids **D.** Glucose	**Ans: B**

Q19. Lactose of milk is formed from: **A.** Two glucose molecules **B.** Glucose and galactose **C.** Fructose and glucose **D.** Two fructose molecules	**Ans: B**

Q20. Glucose is used for the formation of fat and: **A.** Vitamins **B.** Amino acid **C.** Minerals **D.** Hormones	**Ans: B**

Q21. Carbon skeleton of fats, chitin, cellulose is used in: **A.** Respiration only **B.** Formation of fat and amino acid in the body **C.** Blood production **D.** Bone formation	**Ans: B**

Q22. Sources of carbohydrate include: **A.** Meat and fish only **B.** Potato, wheat, rice, maize, sweet potato, and other plant and animal materials **C.** Eggs only **D.** Milk only	**Ans: B**

Q23. Protein was first used by: **A.** Hopkins **B.** J. Berzelius **C.** Funk **D.** Pasteur	**Ans: B**
Q24. Proteins are complex organic compounds made of: **A.** Fatty acids **B.** Amino acids (approximately 16% of the human body is protein) **C.** Nucleic acids **D.** Glucose units	**Ans: B**
Q25. How many types of amino acids are necessary for protein synthesis in the human body? **A.** 10 **B.** 20 **C.** 30 **D.** 15	**Ans: B**
Q26. Out of 20 amino acids, how many are synthesized by the body itself? **A.** 8 **B.** 12 **C.** 15 **D.** 20	**Ans: B**
Q27. Nitrogen is present in protein at approximately: **A.** 10% **B.** 16% **C.** 25% **D.** 5%	**Ans: B**
Q28. Production of energy: By oxidation of pyruvic acid, from one molecule NADH, how many molecules of ATP are formed? **A.** 2 **B.** 3 **C.** 1 **D.** 5	**Ans: B**
Q29. From one molecule of NADH2, how many molecules of ATP are formed? **A.** 2 **B.** 3 **C.** 1 **D.** 5	**Ans: B**

Q30. **Total ATP molecules produced from one molecule of glucose in aerobic respiration is:** **A.** 36 **B.** 38 **C.** 2 **D.** 4	**Ans: B**

Q31. **From glycolysis, how many molecules of NADH are produced from one glucose molecule?** **A.** 4 **B.** 2 **C.** 1 **D.** 6	**Ans: B**

Q32. **Kreb's cycle produces how many ATP molecules per glucose?** **A.** 38 **B.** 24 **C.** 2 **D.** 4	**Ans: B**

Q33. **The main respiratory substances are:** **A.** Only fats **B.** Carbohydrate, fat, and protein **C.** Only minerals **D.** Only vitamins	**Ans: B**

Q34. **The part of eye which is responsible for the conversion of image into neural signal is:** **A.** Cornea **B.** Retina **C.** Iris **D.** Lens	**Ans: B**

Q35. **Retina of eye contains cells called:** **A.** Neurons **B.** Photoreceptor cells **C.** Muscle cells **D.** Blood cells	**Ans: B**

Q36. **The optic nerve carries image information to the:** **A.** Spinal cord **B.** Brain **C.** Heart **D.** Lungs	**Ans: B**

Q37. Cyanide poisoning causes death by affecting: **A.** Digestive system **B.** Electron transport chain system **C.** Skeletal system **D.** Endocrine system	**Ans: B**

Q38. Organisms get energy from: **A.** Only water **B.** Nutrients (mainly carbohydrate and fat used for biosynthesis and energy) **C.** Only air **D.** Only minerals	**Ans: B**

Q39. Assimilation means: **A.** Eating food **B.** Absorption of food and utilization for body functions **C.** Excretion of waste **D.** Breathing	**Ans: B**

Q40. Undigested foods are included in: **A.** Nutrition **B.** Excretion/egestion **C.** Assimilation **D.** Absorption	**Ans: B**

Q41. Monosaccharides containing six carbon atoms are called: **A.** Pentoses **B.** Hexoses **C.** Trioses **D.** Tetroses	**Ans: B**

Q42. Starch is a: **A.** Monosaccharide **B.** Polysaccharide **C.** Disaccharide **D.** Amino acid	**Ans: B**

Q43. Glycogen is the storage form of carbohydrate in: **A.** Plants **B.** Animals **C.** Bacteria **D.** Fungi only	**Ans: B**

Q44. Cellulose is found in: **A.** Animal muscles **B.** Plant cell walls **C.** Blood **D.** Bones	**Ans: B**

Q45. Pentose sugars include: **A.** Glucose and Fructose **B.** Ribose and Deoxyribose **C.** Sucrose and Maltose **D.** Starch and Glycogen	**Ans: B**

Q46. Sucrose is formed by joining: **A.** Two glucose molecules **B.** Glucose and Fructose **C.** Glucose and Galactose **D.** Two fructose molecules	**Ans: B**

Q47. Maltose is formed by joining: **A.** Glucose and fructose **B.** Two glucose molecules **C.** Glucose and galactose **D.** Two fructose molecules	**Ans: B**

Q48. Which polysaccharide forms the exoskeleton of arthropods? **A.** Starch **B.** Chitin **C.** Glycogen **D.** Cellulose	**Ans: B**

Q49. Essential amino acids are those which: **A.** Body can synthesize **B.** Must be obtained from food (8 in number) **C.** Are not needed **D.** Are only in plants	**Ans: B**

Q50. During complete aerobic respiration of one glucose molecule, how many calories of energy is produced? **A.** 1000 kJ **B.** 2870 kJ (686 kcal) **C.** 500 kJ **D.** 100 kJ	**Ans: B**

Answer Key & Explanations

Q1. Answer: B — Nutrition
Page 421: To maintain life, organisms need nutrition. Nutrition is the basic function of food.

Topic: Nutrients

Q2. Answer: B — Carbohydrates, proteins, fats, minerals, vitamins and water
Page 421: Nutrients include carbohydrates, proteins, fats, minerals, vitamins and water.

Topic: Nutrients

Q3. Answer: B — Carbon, Hydrogen and Oxygen
Page 421: Carbohydrates are organic compounds in which the ratio of Carbon, Hydrogen and Oxygen is 1:2:1.

Topic: Carbohydrates

Q4. Answer: B — 50 to 75%
Page 421: Carbohydrate is the major intake providing 50 to 75% energy.

Topic: Carbohydrates

Q5. Answer: B — Aldose
Page 421: Carbohydrate containing aldehyde group is called aldose and with ketone group is called ketose.

Topic: Carbohydrates

Q6. Answer: B — 3 – Monosaccharides, Oligosaccharides, Polysaccharides
Page 421: Carbohydrates are classified into three main groups: Monosaccharides, Oligosaccharides, Polysaccharides.

Topic: Carbohydrates

Q7. Answer: B — Single polyhydroxy or ketone units
Page 421: Monosaccharides are the simplest carbohydrates, single polyhydroxy or ketone units.

Topic: Carbohydrates

Q8. Answer: B — Glucose
Page 421: Most abundant monosaccharide found in nature is glucose. Containing 6 carbon atoms.

Topic: Carbohydrates

Q9. Answer: B — Hexose sugar
Page 421: Glucose is a type of hexose sugar (6 carbon atoms).

Topic: Carbohydrates

Q10. Answer: B — Fructose
Page 421: The sweetest sugar is fructose.

Topic: Carbohydrates

Q11. Answer: B — Oligosaccharides
Page 421: Oligosaccharides are formed when 2 to 10 monosaccharides join together.

Topic: Carbohydrates

Q12. Answer: B — Disaccharides (crystalline, sweet in taste)

Page 421: They are usually crystalline in nature and sweet in test. Maltose, sucrose, lactose are disaccharides.

Topic: Carbohydrates

Q13. Answer: B — 10

Page 421: Polysaccharides are formed when more than 10 monosaccharides join together.

Topic: Carbohydrates

Q14. Answer: B — Starch, glycogen, cellulose, chitin

Page 421: Polysaccharides examples: starch, glycogen, cellulose, chitin etc. They are large, insoluble, tasteless.

Topic: Carbohydrates

Q15. Answer: B — The cell

Page 421: Glucose is the source of immediate energy production in the cell.

Topic: Carbohydrates

Q16. Answer: B — Oxygen

Page 421: Carbohydrate is used as fuel. During respiration, glucose breaks into CO_2 and H_2O with help of oxygen releasing 4.2 kilo calories energy.

Topic: Carbohydrates

Q17. Answer: B — 4.2

Page 421: One gram of glucose gives 4.2 kilo calories energy.

Topic: Carbohydrates

Q18. Answer: B — Nucleosides and nucleotides

Page 421: Nucleic acids are polymers of nucleosides and nucleotides.

Topic: Carbohydrates

Q19. Answer: B — Glucose and galactose

Page 421: Lactose of milk is formed from glucose and galactose.

Topic: Carbohydrates

Q20. Answer: B — Amino acid

Page 421: Glucose is used for the formation of fat and amino acid.

Topic: Carbohydrates

Q21. Answer: B — Formation of fat and amino acid in the body

Page 421: Carbon skeleton of fats, chitin, cellulose etc. is used in the body for formation of structures.

Topic: Carbohydrates

Q22. Answer: B — Potato, wheat, rice, maize, sweet potato, and other plant and animal materials

Page 421: Sources include potato, wheat, rice, maize, sweet potato, and other plant and animal sources.

Topic: Carbohydrates

Q23. Answer: B — J. Berzelius
Page 421: Protein word was first used by J. Berzelius.

Topic: Proteins

Q24. Answer: B — Amino acids (approximately 16% of the human body is protein)
Page 421: Protein is a complex organic compound made up of amino acids. Approximately 16% of human body is protein.

Topic: Proteins

Q25. Answer: B — 20
Page 421: Twenty types of amino acid are necessary for protein synthesis.

Topic: Proteins

Q26. Answer: B — 12
Page 421: Out of 20, 12 are synthesized by body itself and remaining 8 are obtained from food (essential amino acids).

Topic: Proteins

Q27. Answer: B — 16%
Page 421: Nitrogen is 16% present in protein.

Topic: Proteins

Q28. Answer: B — 3
Page 421: By the oxidation of NADH, 3 molecules of ATP are formed.

Topic: Energy Production

Q29. Answer: B — 3
Page 421: One molecule of NADH2 gives 3 ATP molecules.

Topic: Energy Production

Q30. Answer: B — 38
Page 421: ATP glucose total = 2 + 36 = 38 ATP molecules from complete oxidation of glucose.

Topic: Energy Production

Q31. Answer: B — 2
Page 421: From glycolysis, 2 NADH molecules are produced.

Topic: Energy Production

Q32. Answer: B — 24
Page 421: Kreb's cycle produces 24 ATP (from complete cycle for both pyruvate molecules).

Topic: Energy Production

Q33. Answer: B — Carbohydrate, fat, and protein
Page 421: Main respiratory substances are carbohydrate, fat and protein. At first, oxidation of glucose takes place, then fat, then protein.

Topic: Respiration

Q34. Answer: B — Retina

Page 421: The part of eye responsible for conversion of image into neural signal is Retina. It contains photoreceptor cells.

Topic: Eye/Vision

Q35. Answer: B — Photoreceptor cells

Page 421: Retina contains photoreceptor cells and sends information to brain to decide what the picture is.

Topic: Eye/Vision

Q36. Answer: B — Brain

Page 421: Retina side of eye processes and sends information to the brain through optic nerve.

Topic: Eye/Vision

Q37. Answer: B — Electron transport chain system

Page 421: Cyanide poisoning causes death by breaking down the electron transport chain/equalising the pressure on the system.

Topic: Respiration

Q38. Answer: B — Nutrients (mainly carbohydrate and fat used for biosynthesis and energy)

Page 421: Organisms get energy from nutrients. Carbohydrate and fat are used for biosynthesis and energy.

Topic: Nutrients

Q39. Answer: B — Absorption of food and utilization for body functions

Page 421: Assimilation is the absorption of nutrients and their utilization for body functions.

Topic: Nutrients

Q40. Answer: B — Excretion/egestion

Page 421: Function of food includes nutrition, digestion, absorption, assimilation and egestion of undigested foods.

Topic: Nutrients

Q41. Answer: B — Hexoses

Page 421: Monosaccharides with 6 carbon atoms are called hexoses (e.g., glucose, fructose).

Topic: Carbohydrates

Q42. Answer: B — Polysaccharide

Page 421: Starch is a polysaccharide. It is the storage form of carbohydrate in plants.

Topic: Carbohydrates

Q43. Answer: B — Animals

Page 421: Glycogen is the storage form of carbohydrate in animals.

Topic: Carbohydrates

Q44. Answer: B — Plant cell walls

Page 421: Cellulose is found in plant cell walls. It is a structural polysaccharide.

Topic: Carbohydrates

Q45. Answer: B — Ribose and Deoxyribose

Page 421: Monosaccharides with 5 carbons include ribose (in RNA) and deoxyribose (in DNA).

Topic: Carbohydrates

Q46. Answer: B — Glucose and Fructose

Page 421: Sucrose (table sugar) is formed by joining glucose and fructose.

Topic: Carbohydrates

Q47. Answer: B — Two glucose molecules

Page 421: Maltose is formed by joining two glucose molecules.

Topic: Carbohydrates

Q48. Answer: B — Chitin

Page 421: Chitin is the polysaccharide that forms exoskeleton of arthropods.

Topic: Carbohydrates

Q49. Answer: B — Must be obtained from food (8 in number)

Page 421: Essential amino acids are 8 in number and must be obtained from food.

Topic: Proteins

Q50. Answer: B — 2870 kJ (686 kcal)

Page 421: $C6H12O6 + 6O2 \rightarrow 6CO2 + 6H2O$ + 2870 kJ energy.

Topic: Energy Production